

How the fly holds a single goal: normalization, not selection, in *Drosophila* FC2

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Abstract

A walking fly steers toward a goal direction, held as a bump of activity across the FC2 neurons of the fan-shaped body. These neurons also inhibit one another over distance, more strongly the farther apart they are, a feedback proposed to keep the fly on a single goal. We asked, from the connectome, what circuit produces this inhibition, and whether it lets FC2 actively *choose* one goal among competitors (a winner-take-all) or simply keeps a goal set elsewhere as one clean bump. Tracing the wiring in a single FlyWire brain, we find the inhibition is almost entirely global: four FB5A cells inhibit every FC2 neuron roughly equally, with a smaller, distance-dependent contribution from $h\Delta$ interneurons and a negligible direct component. A ring-attractor winner-take-all (the kind the compass uses) requires local recurrent excitation that the FC2 wiring lacks, so this geometry cannot build one; and across a range of dynamical models, including a spiking network, no version of the circuit locks onto a winner at the connectome-scaled reference coupling. FC2 therefore *normalizes* an externally set goal rather than selecting it, with FB5A likely acting as the global normalizer, much as the APL neuron does in the mushroom body. We are explicit about two open points: a different mechanism, mutual inhibition between two competing goals (which $h\Delta$ supplies), could in principle select at very strong coupling, and we bound rather than exclude it; and FB5A’s inhibitory identity is a low-confidence prediction of the connectome’s transmitter classifier, not yet measured, and likely not GABAergic. We then ask where the goal is actually set: the connectome nominates an upstream $h\Delta$ network and rules out the leading proposed alternative, whose neurons supply under 0.2% of FC2’s input. Finally, we propose a direct experiment, silencing FB5A while imaging FC2, that would test the account.

1 Introduction

Navigating animals hold a *goal*: a direction, in world coordinates, they intend to travel. In *Drosophila*, menotaxis, holding a fixed bearing relative to a distant cue, is the paradigmatic case [Giraldo et al., 2018, Green et al., 2019], and the central-complex circuit that implements it is now well characterized. A heading estimate is maintained as a single bump of activity across the EPG “compass” neurons [Seelig and Jayaraman, 2015, Kim et al., 2017]; a goal direction is maintained as a bump across the FC2 neurons of the fan-shaped body [Mussells Pires et al., 2024]; and the PFL3 neurons convert the heading–goal difference into a left–right steering command, while PFL2 neurons, tuned to the anti-goal direction, set the *gain* of that command [Westeinde et al., 2024, Mussells Pires et al., 2024].

Mussells Pires et al. [2024] also reported a suggestive property of the goal bump itself: optogenetically activating FC2 in one fan-shaped-body column *suppresses* FC2 activity in other columns, and *more strongly the farther away*, feedback inhibition they proposed serves to keep a single goal bump. The mediating circuit and its computational character were left open. That is our starting

point. We ask, from the connectome: *what implements this feedback inhibition, and is it the kind of local lateral inhibition that would build a winner-take-all, or something else?*

The intuitive reading, and why it needs testing. The natural reading of “single-bump-enforcing inhibition” is a winner-take-all like the EPG compass, which selects one heading through a continuous ring attractor, *local* recurrent excitation between neighbouring columns, balanced by global inhibition [Kim et al., 2017]. The connectome even offers a clean inhibitory candidate: FB5A, four tangential cells, *predicted* GABAergic by the FlyWire transmitter classifier [Eckstein et al., 2024] (confidence ~ 0.79 ; no experimentally verified transmitter, so we treat the inhibitory identity as a prediction), that contact *all* 85 FC2 neurons (7,741 synapses). But a WTA needs local excitation, and whether the FC2 circuit has it is an empirical question the connectome can answer.

Contributions. (i) We *decompose* the reported distant inhibition into a uniform (FB5A) component, an anti-local ($h\Delta$) component, and a negligible direct component (Sec. 4.1), sharpening Mussells Pires et al. [2024] and the known $h\Delta$ vector-summation motif [Hulse et al., 2021, Lyu et al., 2022] into a cell-type-resolved account. (ii) We show the inhibition is global and anti-local, *not* local, so the circuit cannot host a within-FC2 ring-attractor WTA, a constraint we verify across model families with a bistability test, the wiring replicating in a second connectome (Secs. 4.1 and 4.2); this reconciles the “single-bump” inhibition of Mussells Pires et al. [2024] with the fact that the FC2 wiring is not a ring attractor of the kind now modelled for adjacent fan-shaped-body circuits [Lanz et al., 2025]. (iii) We interpret FB5A as a uniform normalizer, the APL motif implemented by a specific cell (Sec. 4.3). (iv) Because selection is upstream, we ask the connectome *where*: it nominates the $h\Delta$ C-led recurrent network as the goal-holding substrate FC2 reads (valence-gated via FB5AB, a distinct sibling cell type of FB5A), and *excludes* the Lanz $h\Delta$ K–PFG attractor as FC2’s source, since it supplies $<0.2\%$ of FC2’s input (Sec. 4.4). (v) We derive a falsifiable prediction: under two-cue competition, silencing FB5A preserves the relative activation of the competing columns (a global FB5A) rather than changing which dominates (a local selector), with overall disinhibition as a secondary signature (Sec. 4.7). We are explicit throughout about what is earned (the decomposition, the no-local-WTA constraint, and the $h\Delta$ K–PFG exclusion, all transmitter-independent) versus offered (the normalizer role and the goal-substrate nomination, falsifiable candidates we specify direct experiments to test).

2 Related work

The heading–goal–steering axis. The EPG “compass” maintains a single heading bump through a continuous ring attractor, local recurrent excitation balanced by global inhibition [Seelig and Jayaraman, 2015, Kim et al., 2017], the canonical central-complex winner-take-all, and the architecture we ask whether FC2 copies. Downstream, PFL3/PFL2 read the FC2 goal against the compass heading to produce an opponent steering command [Westeinde et al., 2024, Mussells Pires et al., 2024]. Mussells Pires et al. [2024] established the FC2 goal bump and reported the distant feedback inhibition this paper decomposes.

Fan-shaped-body circuitry and goal models. Hulse et al. [2021] reconstructed the central-complex connectome, characterized the $h\Delta$ vector-summation motif (half-fan-shaped-body offset), and framed the fan-shaped body as a context-dependent action-selection centre; Lyu et al. [2022] then demonstrated experimentally that the $h\Delta$ B layer performs this vector summation in behaving

flies. Matheson et al. [2022] showed distributed odour-and-wind cues are integrated within the fan-shaped body (via $h\Delta C$ and tangential gating) into a navigational goal. Lanz et al. [2025] modelled a fan-shaped-body goal circuit ($h\Delta K + PFG$) as a disinhibition-gated recurrent attractor read out by PFL, the strongest persistent-goal mechanism proposed to date (currently a preprint), and the alternative we test FC2 against.

The normalization motif and transmitter caution. Global feedback inhibition that normalizes a population into a sparse code is the motif the mushroom-body APL neuron implements [Lin et al., 2014, Dasgupta et al., 2017]; APL inhibition is itself spatially graded rather than perfectly uniform [Amin et al., 2020]. Transmitter identity in the fan-shaped body is treacherous: dorsal-FB / 23E10 tangential neurons long reported GABAergic were recently shown to express no GABA [Jones et al., 2025], and in the fly olfactory system glutamate is inhibitory via $\text{GluCl}\alpha$ chloride channels [Liu and Wilson, 2013], with GABA receptors (Rdl/GABA-B) present in the central complex [Enell et al., 2007]. These bracket the transmitter-contingent parts of our account.

3 Methods

3.1 Connectome data and the preferred-bearing axis

We source all wiring from a single FlyWire brain [Dorkenwald et al., 2024, Schlegel et al., 2024] so that every connection is a real reconstructed edge rather than an assumed one; predicted transmitters are the FlyWire connectome’s synapse-image neurotransmitter classifier [Eckstein et al., 2024], surfaced per neuron by Schlegel et al. [2024]. The relevant populations are the 85 FC2 neurons, the four FB5A neurons, and the $h\Delta/v\Delta$ interneurons. Each FC2 neuron has a preferred bearing ψ_i recovered from the spectral structure of the FC2/PFL columnar connectivity fingerprint (not from FB5A or $h\Delta$, so bearing-based tests on those cells are not circular). Route weights are disynaptic products profile($W_{\text{FC2} \rightarrow X} W_{X \rightarrow \text{FC2}}$) binned by bearing distance, on a common synapse-count scale. All measurements are repeated in the independently reconstructed hemibrain [Scheffer et al., 2020, Hulse et al., 2021], where the real anatomical column index gives an axis independent of the spectral one.

3.2 The bistability (seeded-basin) test and model families

Operationally, a winner-take-all is *bistable*: which candidate wins depends on the initial state. We drive a model with two competing goals, seed the dynamics toward the left vs. the right candidate, and measure the *basin gap* = (settled bump location | seeded-right) – (settled | seeded-left). A gap $\approx 0^\circ$ means one input-determined fixed point (no WTA); a large gap means history-dependent latching (a WTA). We test, each with connectome-imposed connectivity and a handful of free gains: **(1)** subtractive inhibition $x_i = [\text{drive}_i - \text{global inhibition}]_+$; **(2)** a spiking-proxy (noisy-rate) model, and separately a committed leaky integrate-and-fire network (85 FC2 neurons with threshold/reset/leak and FB5A global-inhibition feedback); **(3)** divisive normalization (Sec. 3.3); **(4)** the measured $\text{FC2} \rightarrow h\Delta \rightarrow \text{FC2}$ antipodal mutual-inhibition matrix; and **(5)** the one *local* route, $v\Delta$, run as local *excitation* (the ingredient a ring-attractor WTA needs). Two controls use an idealized uniform ring: a positive control adds local excitation (should latch), a negative control is global-only (should not). We use a no-latch bound of 30° and a WTA bound of 60° ; the positive control lands at 114° , validating the detector. To exclude non-recurrent selection we additionally sweep the divisive sharpening exponent p (feedforward max-selection).

3.3 The divisive-normalization model

We instantiate the account as divisive normalization [Heeger, 1992, Carandini and Heeger, 2012], a reusable global-inhibition module on the real FC2 ring (a recurrent variant of the canonical form: the sharpening exponent p acts on the feedforward drive in the numerator, while the denominator pools the recurrent population activity). Each FC2 neuron receives a drive; FB5A feeds back inhibition proportional to the pooled population activity, divided across the ring:

$$x_i = \frac{[\text{drive}_i]_+^p}{\sigma + g_{\text{inh}} w_i (\mathbf{pool} \cdot \mathbf{x})}, \quad (1)$$

where drive_i is the feedforward input to FC2 cell i (a von Mises bump centred on the goal bearing, concentration κ), w_i are the real FB5A→FC2 per-cell inhibition weights, $\mathbf{pool}$ the real FC2→FB5A pooling weights (both from FlyWire), and $(p, g_{\text{inh}}, \sigma)$, with σ the semi-saturation constant, are gains set the fly’s way, from a task-agnostic objective (a single clean bump for a single committed goal), *never* from any downstream task. We report the spatial non-uniformity of an inhibition profile as its *modulation*, the peak-to-peak range across bearing bins divided by the mean (so 0 is perfectly uniform); the two-cue sweep parameter α scales an imposed cosine non-uniformity that replaces the measured w_i in the sweep only, $w_i \propto 1 + \alpha \cos \psi_i$. Setting $g_{\text{inh}} = 0$ silences FB5A (the prediction’s control). We read two quantities from two integrations of Eq. 1: a per-step-normalized iteration for the bump’s *shape* (scale-free), and an un-normalized iteration for the settled *amplitude*. Reported concentrations are from the normalized form; the printed equation iterated raw gives a matching 0.88. The feedforward co-representation and two-cue ratio results below use the shipped per-step-normalized form; because both are scale-invariant ratios, they are unchanged by the normalization (we re-ran raw Eq. 1: loser-mass and ratio move by <1%). The bistability test alone omits per-step normalization, which would otherwise clamp the runaway a WTA needs. For robustness we sweep the free gains over a $3 \times 3 \times 3$ grid (power $\times \{0.7, 1.0, 1.4\}$, $g_{\text{inh}} \times \{0.5, 1, 2\}$, $\sigma \times \{0.5, 1, 2\}$ about the committed operating point).

3.4 The two-cue competition test

To distinguish a spatially-uniform FB5A from a within-FC2 local selector, we present two competing goals (two drive bumps 120° apart, amplitude ratio 1:0.7) and compare the competing-column activation ratio with FB5A on vs. silenced, quantified as $|\log(\text{ratio}_{\text{on}}/\text{ratio}_{\text{off}})|$. We repeat this for three uniform functional forms (divisive, subtractive, excitatory-additive) and against a local-excitation selector control, and sweep the spatial non-uniformity α of the inhibition.

3.5 Input tracing and the selector nomination

We classify FC2’s presynaptic types (directional columnar $h\Delta$ /PFN vs. broad FB-tangential), scoring each by synapse mass and per-cell bearing concentration (circular resultant of a cell’s FC2 targets), and trace two-hop routes (MBON→FB5A→ $\{h\Delta\text{C}, \text{FC2}\}$; $h\Delta\text{K}$ /PFG in/out of FC2). Behaviour anchors are measured signatures from FC2 imaging [Mussells Pires et al., 2024] plus one flagged cross-system analogy [Toepfer et al., 2018], not fits.

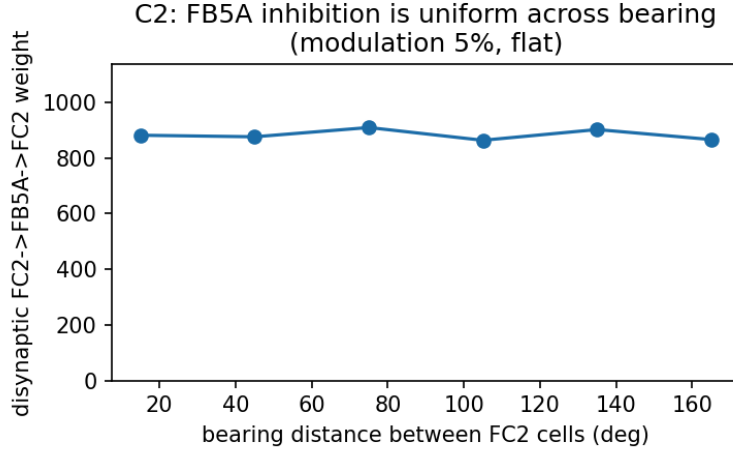


Figure 1: **C2, FB5A inhibition is uniform across bearing.** The disynaptic $\text{FC2} \rightarrow \text{FB5A} \rightarrow \text{FC2}$ loop weight is flat as a function of the bearing distance between FC2 cells (modulation 5%): FB5A is a global scaler, not a spatial selector. Measured from the FlyWire connectome (`analyze_fc2_scoping.py`).

4 Results

4.1 The feedback inhibition decomposes into a uniform, an anti-local, and a negligible term

The reported distant inhibition could arise through three routes, direct $\text{FC2} \rightarrow \text{FC2}$ synapses, the tangential FB5A pool, or the $h\Delta/v\Delta$ interneurons, with very different computational meanings. We measure each.

FB5A is a global, not spatial, inhibitor. If FB5A were the inhibitory arm of a WTA, its inhibition of an FC2 neuron would depend on that neuron’s *bearing distance* from the active bump. It does not (Fig. 1): the four FB5A cells reach all 85 FC2 (85/85); the per-cell $\text{FB5A} \rightarrow \text{FC2}$ weight is uncorrelated with bearing (correlation with $\cos \psi / \sin \psi$ of $+0.02 / -0.04$); and the full disynaptic $\text{FC2} \rightarrow \text{FB5A} \rightarrow \text{FC2}$ loop is *flat* across bearing distance (modulation 5%). FB5A scales the whole population; it does not sculpt it spatially.

The FC2 ring lacks local recurrent excitation. A WTA additionally requires local positive feedback. We find no local recurrent *excitation* in the FC2 wiring (Fig. 2):

- $\text{FC2} \rightarrow h\Delta \rightarrow \text{FC2}$ recurrence (sign undetermined; measured as connectivity geometry, not transmitter) is *anti-local*: it couples bearings $\sim 180^\circ$ apart (a vector-summation motif; disynaptic weight ~ 38 near, dipping mid-range, peaking at ~ 95 antipodally, far/near ratio 2.5), not neighbours.
- $v\Delta$ -mediated recurrence is spatially local but weak (5% of $h\Delta$) and of undetermined sign.
- Direct $\text{FC2} \rightarrow \text{FC2}$ connectivity is negligible.

This geometry, global and anti-local inhibition, with no local recurrent excitation, is not the wiring of a winner-take-all. Whether the circuit can nonetheless latch a choice we settle dynamically in Sec. 4.2 (Fig. 4).

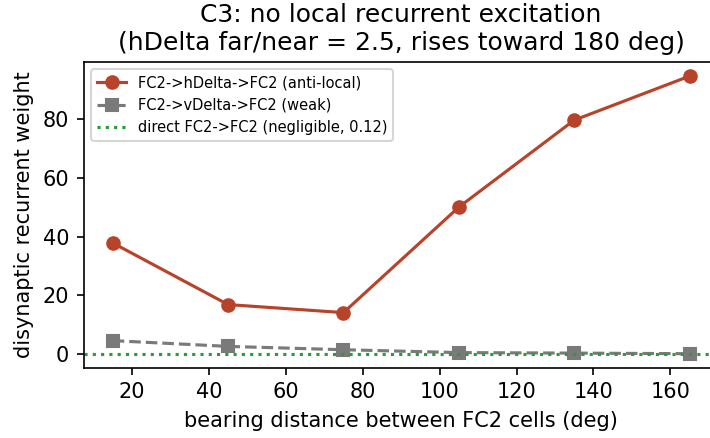


Figure 2: **C3, no local recurrent excitation.** Direct $\text{FC2} \rightarrow \text{FC2}$ is negligible; the $\text{FC2} \rightarrow h\Delta \rightarrow \text{FC2}$ disynaptic weight *peaks at antipodal bearings* (anti-local, near 37.8 $\rightarrow$ far 94.7, far/near 2.5, dipping mid-range; a vector-summation motif, not neighbour coupling); $v\Delta$ is local but weak. The local positive feedback a WTA needs is absent.

What the three routes sum to. On the common disynaptic scale, the total $\text{FC2} \rightarrow \text{FC2}$ inhibitory pathway is a *large distance-uniform component plus a small anti-local one*: FB5A supplies $\sim 95\%$ of the total inhibitory mass but is flat across bearing (far/near 1.0, modulation 5%), contributing no distance-dependence; the $h\Delta$ route supplies only $\sim 5\%$ of the mass yet carries essentially *all* of the distance-dependence ($\sim 85\%$ of the total’s cross-bin variance, a covariance-adjusted attribution; far/near 2.5 in isolation). So the connectome predicts inhibition of the sign Mussells Pires reported, increasing from near to far bearings, but as a modest anti-local rise on a uniform floor rather than a steep gradient; and because a normalizer subtracts the uniform FB5A term (Sec. 4.3), it is the $h\Delta$ component that survives functionally. Two caveats: synapse count is a structural proxy for functional inhibition, and we compare shape against Mussells Pires’ reported sign, not their digitized curve (which we do not have), a decomposition, not a numerical fit.

The structure replicates in a second connectome. In the independently reconstructed hemibrain (Fig. 3), both structural facts hold: FB5A reaches all FC2 (88/88), the disynaptic loop is flat (modulation 1.2%), direct $\text{FC2} \rightarrow \text{FC2}$ is negligible, and $\text{FC2} \rightarrow h\Delta \rightarrow \text{FC2}$ is anti-local with far/near ratio 2.43, matching FlyWire’s 2.50 (at full precision 2.429 hemibrain vs. 2.504 FlyWire, a 3% difference), across two brains reconstructed by different groups. (The hemibrain is a partial volume, so the raw per-cell FB5A weight carries a completeness gradient that tracks per-column total synapses at $r=0.93$; the bearing-uniformity check therefore uses the completeness-normalized weight (which drops the residual bearing correlation from a raw 0.35 to 0.145), and the gradient is reported, not hidden.)

4.2 No within-FC2 ring-attractor winner-take-all

We asked whether *any* plausible dynamical model of the $\text{FC2} + \text{FB5A}$ circuit, parameterized by the real connectome, collapses competing goals to a single winner (Sec. 3.2). None commits. Divisive normalization *shifts* the bump toward the stronger candidate (selection by salience) but does *not* collapse to a single winner, not even at 20% amplitude asymmetry. A seeded-basin test finds no

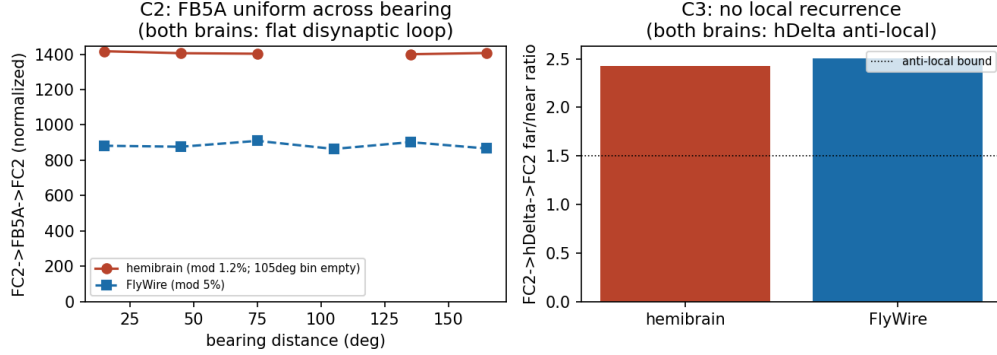


Figure 3: **Two brains agree.** The FB5A-uniformity (left) and the $h\Delta$ anti-local recurrence (right) reproduce in both the FlyWire and hemibrain connectomes (`analyze_fc2_scoping_hemibrain.py`). The dotted line on the right marks the anti-local threshold (far/near ratio 1.5). The hemibrain’s 105° bearing bin is empty (partial-volume dataset), hence the gap in that curve.

bistability in any connectome-parameterized family (basin gap $\approx 24^\circ$ or less at the reference gains, below the 30° no-latch bound and far short of the 114° bistable control): no family latches within the no-latch bound ($h\Delta$, at 24° , is closest). The spiking-proxy also fails, as does a committed LIF spiking network (basin gap 0°); absent local recurrent excitation there is no ring-attractor for spike timing to latch into. The structural core, the absence of the local recurrent excitation a ring-attractor WTA needs (Sec. 4.1), holds across formalisms: **there is no within-FC2 ring-attractor winner-take-all in the connectome** (Fig. 4); the $h\Delta$ mutual-inhibition route is the one gain-contingent exception, confronted below.

No feedforward max-selection either. A bistability test rules out a *recurrent* WTA, but a divisive normalization with a high enough exponent can approach a hard max-selector without any recurrence. Sweeping the sharpening exponent p (Fig. 5), at the task-agnostic operating point ($p=2$) the circuit *co-represents* competitors: two equal goals leave both bumps (45%/55%), and the stronger of two unequal goals leads with the weaker retained (60%/40%). Collapse to a single column appears only at $p \gtrsim 12$, six-fold above the operating point, and even then it is *not* a winner-take-all: it is init-independent (seed-spread 0.00%, as expected for an input-clamped map), so it amplifies a fixed anatomical bias rather than a history-dependent choice; and it does *not* track the goal, at high p the *stronger* input loses (46% at $p=30$). The feedforward route yields no goal-WTA at any exponent, confirming the bistability result from an independent axis.

Reconciling “single-bump inhibition” with “no winner-take-all”. This is not a contradiction of Mussells Pires et al. [2024]: their distant inhibition is exactly what our decomposition attributes to the uniform FB5A and anti-local $h\Delta$ terms. But this geometry, global plus anti-local, no local excitation, is not the machinery of a WTA. The FC2 circuit has a single input-determined fixed point; it sharpens and normalizes the input but cannot latch a self-sustaining choice.

The $h\Delta$ route, confronted. A skeptic will press the $h\Delta$ family, and rightly: a two-goal winner-take-all needs only mutual inhibition between the competitors, not a continuous ring attractor, and $h\Delta$ supplies exactly that between antipodal columns. Worse, the behaviour we anchor to, two goals 180° apart yield a committed choice, not an average (Sec. 4.6), is the *antipodal* configuration

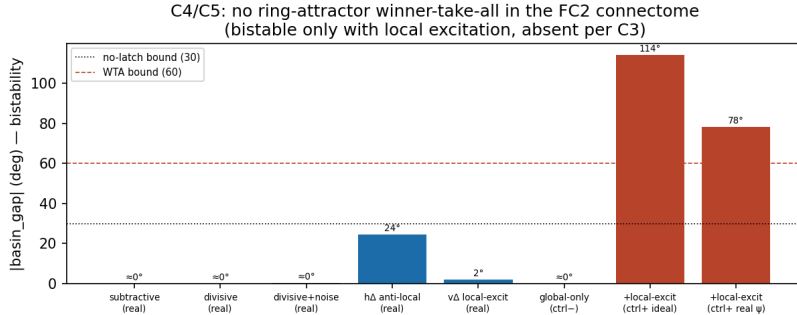


Figure 4: **C4/C5, no bistability, hence no WTA.** Seeded-basin test: the pure global-inhibition families (subtractive, divisive, divisive+noise), the one local route $v\Delta$ run as local excitation (2°), and a global-only control ring have $|\text{basin gap}| \approx 0^\circ$ (one input-determined fixed point). The $h\Delta$ anti-local mutual-inhibition family sits at 24° at the reference gain, below the 30° no-latch bound but the one gain-contingent case (it crosses that bound above gain ≈ 6 , peaking at 38.4° , a third of the 114° positive control; see text). Only the positive controls, local excitation added to the idealized ring (114°) and to the real irregular bearings (78°), are both bistable, confirming the test detects a WTA when one exists, on the real geometry too (so the negative results are not an artifact of irregular sampling suppressing detection), and identifying local excitation, which the real FC2 ring structurally lacks (C3), as the ingredient that would be needed (`prove_fc2_no_wta.py`; the committed leaky integrate-and-fire run, basin gap 0° , is reported in the text and `analyze_fc2.lif.py` but not drawn as a bar here).

where $h\Delta$ coupling peaks. So we state the alternative plainly: the $\text{FC2} \leftrightarrow h\Delta$ loop is the leading candidate for an *in-FC2* two-goal selector, and our result does not exclude it, it bounds it. At the connectome-scaled reference gain its basin gap is 24° (below the 30° no-latch bound); sweeping the anatomically-unconstrained scalar gain, it crosses that bound above gain ≈ 6 (32°) and peaks at 38.4° (gain 8), but *never* reaches the 60° WTA bound at any gain we tested (and is non-monotonic beyond, 32° at gain 10). So the honest statement is intermediate: $h\Delta$ shows a partial, gain-dependent seed-dependence but does not reach full winner-take-all in the tested range, and the 30° bound is a chosen threshold, not a measured one. What does *not* depend on any of this: the pure global-inhibition families do not latch at any gain, and the one local route, $v\Delta$, does not latch *even* run as local excitation and swept over the same gain range as $h\Delta$ (basin gap $\leq 2^\circ$ throughout, never crossing the no-latch bound, unlike $h\Delta$), so “no local excitation” is a gain-robust tested result, not a sign assumption or a claim only about the reference gain. Whether the biological $h\Delta$ gain sits in the latching regime is the single measurement that would decide between our normalization account and an $h\Delta$ selector, and it is exactly what silencing or scaling $h\Delta$ during two-goal imaging would test. We flag this as the primary open alternative, not a closed case.

4.3 FB5A as a uniform normalizer, distributedly sourced

If FB5A is not a spatial selector, the natural reading of a uniform inhibitory pool is **divisive normalization** (a soft k -WTA): it scales the population by its total activity, so the strongest drive survives best and a single clean bump is favoured, without ever being *forced*. This is the motif the mushroom body uses: the APL neuron provides feedback inhibition to the Kenyon cells, producing a sparse, normalized code [Lin et al., 2014, Dasgupta et al., 2017]. Computationally, and *if FB5A is inhibitory*, FB5A:FC2 and APL:KC play the same role. This is an instance of a

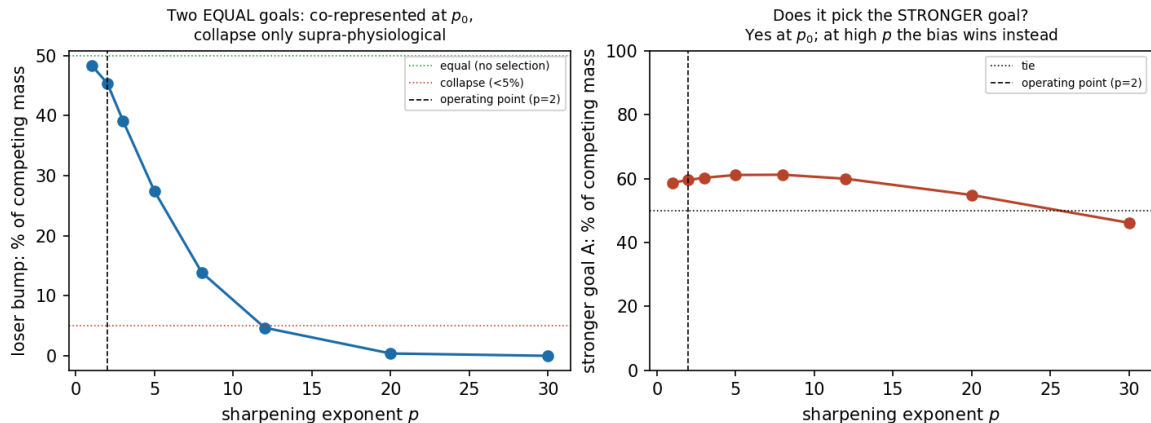


Figure 5: **No winner-take-all via the feedforward sharpening term.** Left: with two *equal* competing goals, the loser bump keeps $\sim$ half the mass at the operating point ($p=2$) and collapses only supra-physiologically ($p \gtrsim 12$); that collapse is init-independent (a fixed bias, not a bistable choice). Right: the sharpening picks the *stronger* goal at p_0 but at high p the fixed anatomical bias wins instead (stronger input drops below 50%), so it never functions as a goal-selector (`analyze_fc2_feedforward.py`).

canonical motif, not a new principle: the contribution is the cell-type assignment. We use “global” in the computational sense; anatomically even APL inhibition is spatially graded [Amin et al., 2020], so the shared identity is the *normalization computation*.

If FC2 does not itself select, the choice is shaped by its inputs, consistent with the fan-shaped body as a context-dependent action-selection centre [Hulse et al., 2021] integrating distributed drives into a goal [Matheson et al., 2022]. Our measurements localize the selector *outside* the FC2 ring. Tracing FC2’s inputs (Fig. 6): FC2 is fed (100% typed) by FB-tangential neurons (55%; a broad family including FB5A), $h\Delta/v\Delta$ interneurons (37%), and PFN neurons (5%; $\sim$ 3% other); there is *no* direct input from mushroom-body output neurons (MBONs contribute $\approx$ 0% directly, so learned valence reaches FC2 indirectly). At single-cell resolution these split into two geometries: each PFN cell targets a *narrow* bearing band ($\sim$ 10 of 85 FC2, concentration 0.96), while each FB5A cell blankets nearly the whole population (median 84/85 per cell; all four collectively 85/85, concentration 0.21).

The uniform-inhibition substrate is distributed, not a one-cell bet. Because FB5A’s inhibitory identity is a low-confidence prediction (Sec. 6), we asked which cells supply FC2’s *uniform* drive, and of what transmitter (Fig. 7). FB5A is the *largest* single uniform input (7,741 synapses, 85/85) and the *only* GABA-predicted one, so naming it is not arbitrary, but only $\sim$ 12% of the uniform-input mass. The rest is a large family of FB-tangential cells that is predominantly *glutamatergic* by type count (40 types; $\sim$ 39% of the uniform-input mass), a transmitter that can be inhibitory via GluCl α [Liu and Wilson, 2013]; GABA and glutamate together make $\sim$ 51% of the uniform-input mass inhibitory-*capable*. This is a capability, not a confirmed sign: the GABA leg has receptor plausibility from central-complex Rdl/GABA-B expression [Enell et al., 2007], but the larger glutamate leg requires GluCl α on FC2, which we have not verified. Dorsal fan-shaped-body tangential cells can moreover *co-transmit* glutamate and acetylcholine [Jones et al., 2025], so a predicted-glutamatergic cell is not guaranteed to be net inhibitory. Two consequences: first, the *inhibitory role is robust to FB5A’s specific transmitter*, whether FB5A is GABAergic or (like

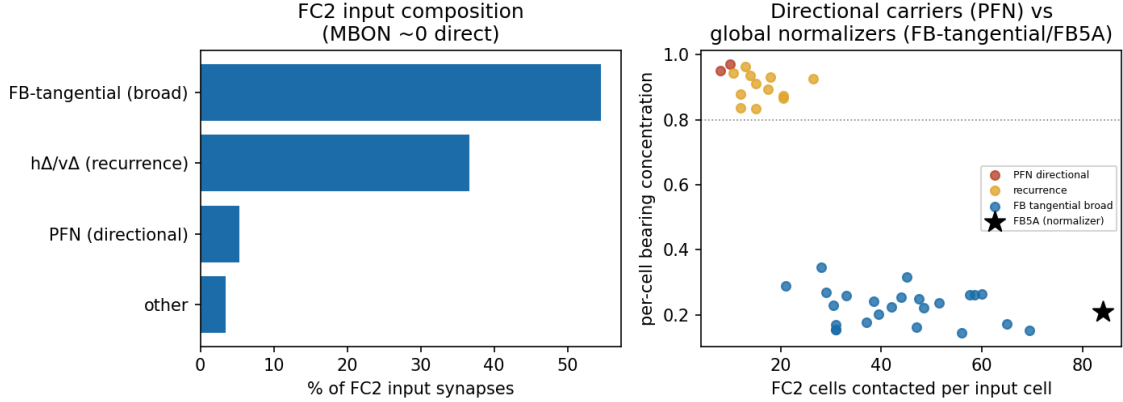


Figure 6: **FC2’s inputs: composition and per-cell geometry.** Left: input composition (FB-tangential dominant; MBON ≈ 0 direct). Right: per-cell targeting, PFN cells are narrow (few FC2 targets), FB5A/FB-tangential cells are broad (bearing-uniform), the geometry a normalizer would use (`trace_fc2_inputs.py`). The direct PFN $\rightarrow$ FC2 edge is minor; the canonical PFN route into FC2 is via $h\Delta$.

Table 1: The four FB5A/ $h\Delta$ input-and-inhibition fractions most easily confused, with their explicit denominators (each individually correct). Rows 1–2 use different denominators from each other and from rows 3–4, so are not comparable; rows 3 and 4 share the same denominator (FC2 total input) and are *nested*, the 28% directional $h\Delta$ is the columnar subset of the 37% recurrent mass. Route fractions elsewhere (e.g. the MBON $\rightarrow$ FB5AB valence path) are quoted as a percentage of the target cell’s input.

Quantity	Of what (denominator)	Where
FB5A $\sim 95\%$	total disynaptic FC2 $\rightarrow$ FC2 <i>inhibitory mass</i>	decomposition sum
FB5A $\sim 12\%$	FC2’s <i>uniform-input</i> synapse mass ($\geq 50\%$ coverage)	§4.3
$h\Delta/v\Delta \sim 37\%$	FC2’s <i>total input</i> synapses (all recurrence)	§4.3
directional $h\Delta \sim 28\%$	FC2’s <i>total input</i> synapses ($h\Delta$ only)	§4.4

sibling FB5AA) glutamatergic, it is inhibitory-capable (the normalization feedback loop we model is established only for FB5A itself, via its FC2 $\rightarrow$ FB5A $\rightarrow$ FC2 pooling; the broad family shows the uniform inhibitory *input* is not one cell’s alone, not that they all form the loop). Second, a caveat the wiring forces: the uniform input is $\sim$ half inhibitory-capable and $\sim$ half cholinergic (excitatory), so the connectome alone cannot declare the *net* uniform drive inhibitory. The earned claim is thus the *computation* plus a *ranked candidate substrate*, with FB5A its leading member; the experiment of Sec. 4.7 decides the single-cell assignment.

4.4 Where the selector is: a connectome-nominated candidate

Having placed goal selection *outside* FC2, we ask the connectome to name the substrate that holds it (Table 1 lists the input fractions used here and their denominators). FC2’s input splits into a broad, non-directional FB-tangential stream ($\sim 55\%$, per-cell concentration ~ 0.2 , which cannot specify a bearing) and a structured columnar stream that can: the recurrent $h\Delta$ network ($\sim 28\%$ of FC2 input, concentration ~ 0.9) and a smaller feedforward PFN vector input ($\sim 5\%$, concentration 0.96) (Fig. 8). Of these, only the $h\Delta$ network is at once *directional*, *high-drive*, and *recurrent* (within-class

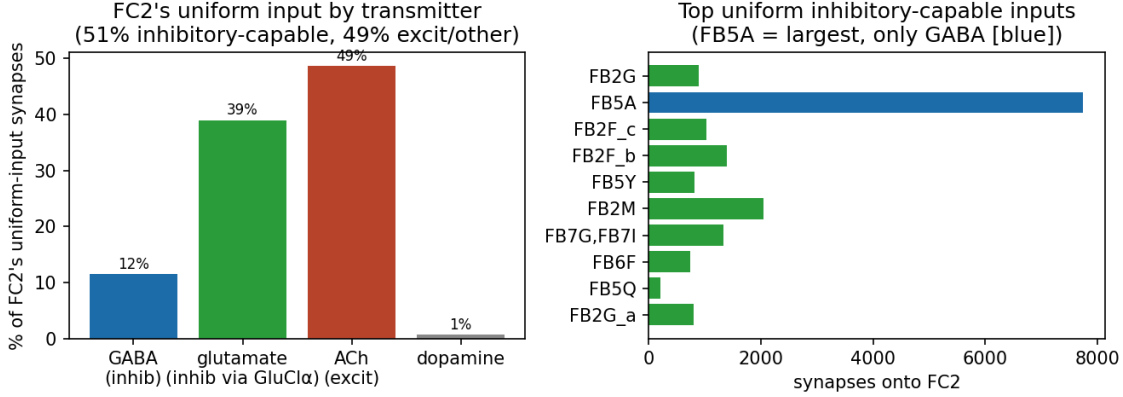


Figure 7: **Who supplies FC2’s uniform input.** Left: FC2’s uniform-input synapse mass by predicted transmitter, $\sim 51\%$ inhibitory-capable (GABA + glutamate/GluCl α), $\sim 49\%$ cholinergic/other (excitatory). Right: the top uniform inhibitory-capable inputs by synapse count, FB5A (blue) is the largest and the only GABA-predicted one, atop a broad glutamatergic FB-tangential family (green) (`analyze_fc2.uniform_inhibitors.py`).

recurrence $\sim 10\%$, the half-fan-shaped-body-offset motif of a vector memory [Hulse et al., 2021]). It is also the only directional input that carries a learned signal: the sole route from mushroom-body output into FC2’s directional substrate is MBON $\rightarrow$ FB5AB $\rightarrow$ $\{h\Delta C, FC2\}$ (MBON $\rightarrow$ FB5AB 5.1%; FB5AB $\rightarrow$ $h\Delta C$ 6.9%, its top named input; FB5AB $\rightarrow$ FC2 1.7%; MBON $\rightarrow$ FC2 direct 0%, each as a percentage of the target cell’s input), and FB5AB is a tangential neuron *positioned* to gate an odour-and-wind goal heading in $h\Delta C$ [Matheson et al., 2022]. We rest the nomination on the *anatomical* MBON $\rightarrow$ FB5AB $\rightarrow$ $h\Delta C$ connectivity above rather than on matching physiology, because the relevant $h\Delta C$ driver line is confounded: VT062617, used to record and perturb $h\Delta C$, was subsequently reported to label the adjacent $h\Delta K$ rather than $h\Delta C$ [Matheson et al., 2024], leaving those functional results cell-type ambiguous (FB5AB itself is targeted by a separate line, 21D07). The anatomical $h\Delta K$ exclusion from FC2 is unaffected by this line ambiguity. We therefore nominate the **$h\Delta$ recurrent network, led by $h\Delta C$, with $h\Delta J$ the largest single $h\Delta$ input to FC2 (6.1%), as the substrate that holds the committed goal FC2 reads**, set by distributed directional and valence-gated drive.

A decisive exclusion. The nomination separates FC2 from the best-modelled alternative. The disinhibition-gated $h\Delta K$ –PFG recurrent attractor [Lanz et al., 2025] is the strongest proposed persistent-goal mechanism in the fan-shaped body, but its readout is PFL, not FC2. The connectome adjudicates cleanly: $h\Delta K$ and PFGs together supply $< 0.2\%$ of FC2’s input (54 and 88 of 78,193 synapses; 0.07% and 0.11%), and $h\Delta K$ projects to PFGs (41% of its output), FB6A and ExR3, only 0.4% to FC2. The $h\Delta K$ –PFG attractor is thus a *parallel, PFL-facing* goal module, not the source of the FC2 bump. This yields a double dissociation: setting or clamping the FC2 goal should require the $h\Delta C$ /FB5AB pathway and be insensitive to $h\Delta K$ /PFG manipulation, whereas the PFL-side persistence of Lanz et al. [2025] should show the opposite dependence.

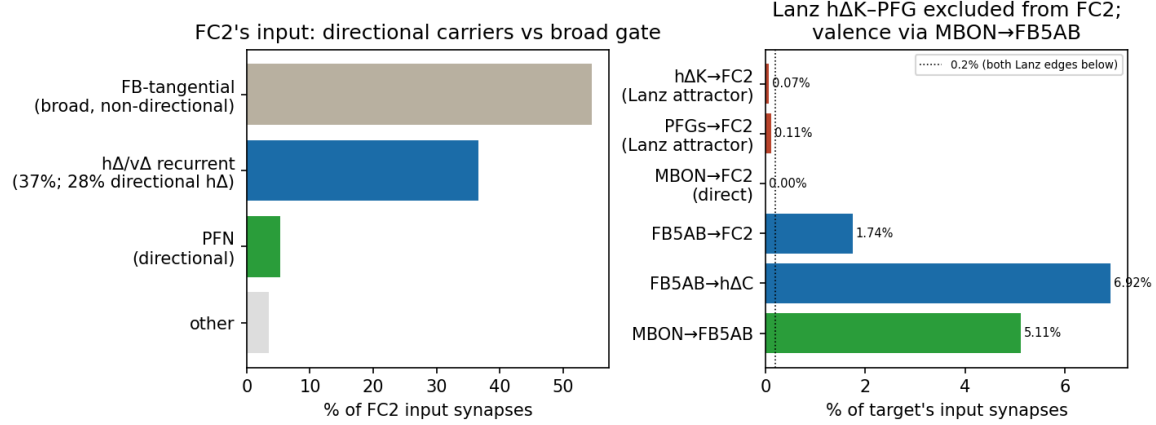


Figure 8: **Where FC2’s goal is set.** Left: FC2’s input is dominated by a broad, non-directional FB-tangential gate ($\sim 55\%$); the directional carriers are the recurrent $h\Delta$ network and the narrow PFN stream. Right: the Lanz $h\Delta$ K-PFG persistent-goal attractor is excluded from FC2 (both $< 0.2\%$), while learned valence enters via $MBON \rightarrow FB5AB \rightarrow \{h\Delta C, FC2\}$ (`analyze_fc2_selector.py`).

4.5 In-silico demonstration

Result 1, the single-goal bump shape is FB5A-invariant. That a single committed goal yields one clean bump (resultant concentration 0.86; 0.88 raw) is how the operating point was chosen (Sec. 3.3), so it is not itself a prediction. The load-bearing observation is that silencing FB5A leaves that concentration near-unchanged ($0.86 \rightarrow 0.85$): the bump’s *shape* is set by the drive and the supralinear term, not by FB5A, the premise the silencing prediction (Result 2) builds on.

Result 2, what silencing FB5A does, and what is robust. We ran the silencing control ($g_{inh} = 0$) and a $3 \times 3 \times 3$ gain sweep. Two candidate effects behave very differently (Fig. 9). (i) A *shape* effect, competing candidates re-splitting the bump, appears at the reference gains but holds in only 30% of the sweep: a threshold artifact we do *not* rely on (uniform inhibition cannot reshape a bump; the supralinear $[\cdot]^p$ term does the sharpening). (ii) An *amplitude* effect: silencing FB5A disinhibits FC2. That activity *rises* is definitional (removing an additive denominator term); the non-trivial measurement is that the bump’s angular tuning is *preserved* while activity rises (median Δ concentration 0.028 across 100% of the sweep). Disinhibition *without reshaping* is the robust signature; the *magnitude* of the rise in this un-normalized limit is a model artifact and we make no magnitude claim.

4.6 Anchoring to fly choice behaviour

The account must match what flies do when they choose. We anchor to four behavioural signatures, three measured in FC2, the fourth an analogy (flagged) [Mussells Pires et al., 2024, Toepfer et al., 2018]:

- The FC2 bump **tracks the goal** (bump–goal correlation $r=0.61$; the EPG heading bump tracks its cue more tightly, 0.88).
- The FC2 bump–goal offset stays **clustered around zero under $\pm 90^\circ$ virtual rotation** (offset distribution significantly directional; V-test $P=6.65 \times 10^{-4}$), the bump tracks an allo-centric goal, not a stimulus-locked response.

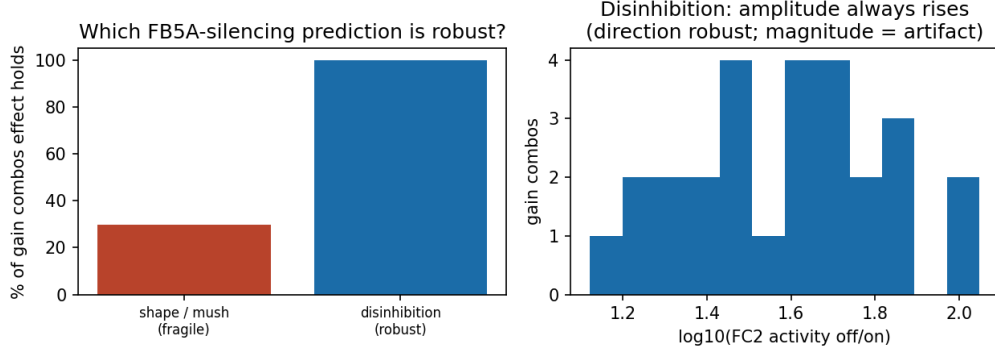


Figure 9: **Which FB5A-silencing effect is robust.** Left: across a $3 \times 3 \times 3$ gain sweep, the shape/“mush” effect holds in only 30% of combinations (fragile), while the disinhibition (amplitude-rise) effect holds in 100%. Right: the direction of the amplitude effect is always the same (activity rises); its magnitude is a model artifact. The paper’s prediction is the robust one (`analyze_fc2_prediction.py`).

- Two goal bumps 180° apart yield two headings $\sim 166^\circ$ apart, the fly **commits**, it does not average.
- Choice is **multi-stable**, stochastically switching between WTA-like and averaging-like regimes, consistent with a soft (normalizing) rather than hard competition. *This last anchor is an analogy to multistable orientation behaviour under ambiguous motion [Toepfer et al., 2018], hence a prediction, not a direct FC2 fit.*

These behaviours are consistent with a normalized, externally-set goal bump, and none requires a within-FC2 winner-take-all, in line with Sec. 4.2. (The commitment signature is also the antipodal configuration that presses the $h\Delta$ alternative; see Sec. 4.2.)

4.7 A falsifiable prediction

The experiment is direct: silence FB5A while imaging FC2 under two competing goals. It is a *single* experiment with a *two-axis* readout forming a four-row discriminator (three decisive outcomes plus a grey-zone intermediate) in which every outcome is informative (reading the amplitude axis first, then the ratio):

Amplitude (<i>total FC2 activity</i>)	Ratio (<i>competing-column ratio</i>)	Verdict
rises	preserved	global inhibitory FB5A, normalizer or uniform bystander (same signature; C2 favours normalizer)
rises	changes which column dominates	a within-FC2 local selector (would overturn C2)
flat / falls	any	FB5A is not inhibitory (transmitter prediction wrong, cf. Jones et al., 2025)
rises	distorted but not flipped	partially non-uniform inhibition (grey zone)

The **ratio** axis is the *locality* test; the **amplitude** axis is the *mechanism* test.

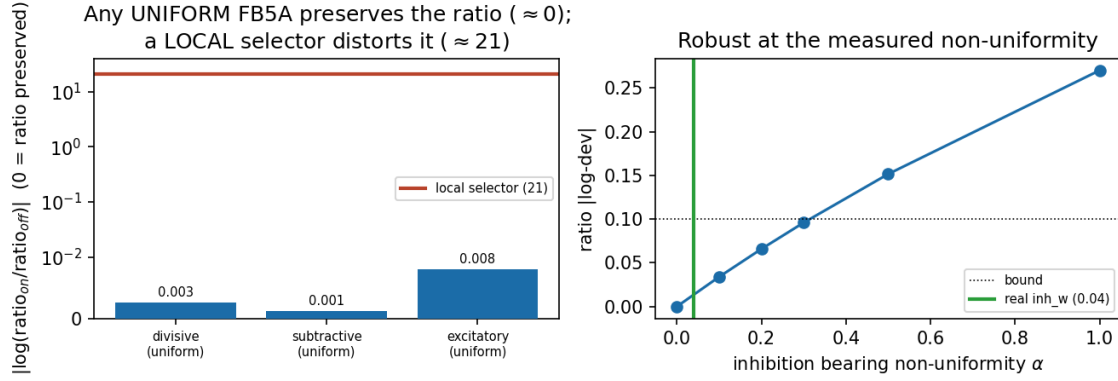


Figure 10: **The two-cue test distinguishes a uniform FB5A from a local selector.** Left: under FB5A silencing, the competing-column ratio is preserved for every *uniform* functional form (divisive/subtractive/excitatory, $|\log\text{-dev}| < 0.01$) but distorted by a local-excitation selector control (≈ 21). Right: the ratio survives inhibition non-uniformity up to $\alpha \approx 0.3$; the measured FB5A non-uniformity (0.04) sits well inside the preserved regime (`analyze_fc2_competition.py`).

Locality axis, competition structure is preserved.

Under two competing goals, silencing FB5A preserves the *relative* activation of the competing FC2 columns (the same ratio, scaled up), whereas a within-FC2 local spatial selector would change which column dominates.

Because a spatially-uniform effect changes both columns nearly in proportion, silencing it leaves the column ratio nearly unchanged, and this holds across the three uniform functional forms we tested, independent of FB5A’s sign (divisive, subtractive, excitatory-additive, the last the Jones et al. [2025] failure mode), all preserving the ratio to $|\log(\text{ratio}_{\text{on}}/\text{ratio}_{\text{off}})| < 0.01$ versus orders of magnitude larger for a local selector ($|\log| \approx 21$, floor-limited; a >2000 -fold gap in the log-deviation metric) (Fig. 10). A preserved ratio is the *expected* outcome given C2 (a uniform FB5A cannot flip the ratio), so this axis *confirms* the measured connectivity and would only surprise us if a within-FC2 local selector existed after all; the mechanism claim is carried by the amplitude axis. The ratio survives inhibition non-uniformity up to $\alpha \approx 0.3$, and the measured FB5A non-uniformity is 0.04, well within margin. Preservation demonstrates FB5A acts *globally* (consistent with a normalizer, but also with uniform gating or a bystander) and *rules out* a within-FC2 local selector; it does not by itself prove the normalizer role. Three wet-lab confounds bear on the measurement, the first a genuine threat: because the fly *commits* to one goal (Sec. 4.6), there may be no stable two-column ratio to read at all, and our workaround (read the transient pre-commitment window, or impose sustained dual cues) is itself an assumption to validate; disinhibition-driven saturation can compress the ratio; and silencing FB5A leaves the anti-local $h\Delta$ term intact.

Amplitude axis, disinhibition without reshaping (the mechanism test). This axis discriminates an *inhibitory* FB5A from a non-inhibitory one, the distinction the locality axis cannot make (it does not separate a normalizer from a uniform inhibitory bystander; C2 does that): silencing an inhibitory, normalizing FB5A should *disinhibit* FC2, raising overall bump activity while preserving angular tuning, rather than reshaping the goal (Fig. 9). We do not lead the whole prediction with this axis because, unlike the locality axis, it *depends on FB5A being inhibitory*; a

flat or falling amplitude would instead be evidence that the predicted GABAergic identity is wrong (the third table row, the Jones et al. [2025] scenario). That contingency is a feature of the joint readout: each decisive outcome tells us something.

5 Discussion

What this paper establishes. (i) A cell-type-resolved **decomposition** of the distant feedback inhibition reported by Mussells Pires et al. [2024]: a uniform FB5A term, an anti-local $h\Delta$ term, a negligible direct term. (ii) A tested **constraint**: this geometry cannot host a within-FC2 ring-attractor winner-take-all, verified across model families with a bistability test, the wiring replicating in a second connectome; the $h\Delta$ mutual-inhibition route (partial gain-dependent seed-dependence, never reaching full WTA) is flagged, not hidden. (iii) A **connectome-nominated upstream substrate** for the goal FC2 reads (the $h\Delta$ C-led recurrent network, valence-gated via FB5AB) and a **decisive exclusion**, the Lanz $h\Delta$ K–PFG attractor supplies <0.2% of FC2’s input, so it is PFL-facing, not FC2’s source. (iv) A **falsifiable prediction** that turns on the measured uniformity, FB5A silencing preserves competition structure (a global FB5A) rather than changing which column dominates (a local selector), plus a disinhibition corollary.

Interpretation. Together these place FC2 as a *normalizer of an externally-set goal*, not its selector: the goal is written into an upstream $h\Delta$ substrate (directional, recurrent, valence-gated) and read out by PFL, while FB5A-type global inhibition keeps the FC2 representation a single clean bump. This reconciles the “single-bump” inhibition of Mussells Pires et al. [2024] with the absence of a ring attractor, and separates FC2’s goal from the parallel $h\Delta$ K–PFG attractor [Lanz et al., 2025]. The novel, transmitter-independent pieces are the decomposition, the no-WTA constraint, and the exclusion; the normalizer role and the substrate nomination are the offered, falsifiable interpretations.

6 Limitations

Prediction, not confirmation. No recordings of the FC2 selection mechanism exist, and the underlying distant-inhibition *phenomenon* is not ours, it was reported by Mussells Pires et al. [2024]; our contribution is its connectome decomposition and computational character.

The transmitter is a low-confidence prediction. FlyWire’s classifier labels FB5A GABAergic at only ~ 0.79 confidence, with no experimental verification, and the evidence is unfavourable: one sibling type is experimentally *GABA-negative* (FB5AB, cholinergic [Matheson et al., 2022]), the other predicted-glutamatergic (FB5AA), and FB-tangential neurons are predominantly glutamatergic, of the experimentally-typed FB-tangential cells in FlyWire, 14 are GABAergic versus 371 GABA-negative. A direct precedent underlines the risk [Jones et al., 2025]. We therefore treat FB5A’s identity as *likely non-GABAergic on balance*, and rest the interpretation on the weaker property of being *inhibitory*: reachable via GABA (Rdl/GABA-B in the central complex [Enell et al., 2007]) or glutamate via GluCl α [Liu and Wilson, 2013]. Neither route is verified on FC2, the GABA route needs FB5A GABAergic (judged unlikely), the glutamate route needs GluCl α on FC2 (unconfirmed), so the inhibitory interpretation is a genuine open question; it fails outright only if FB5A is cholinergic (though the sign-independent locality prediction still survives even then), and is unconfirmed if FB5A is glutamatergic and FC2 lacks GluCl α . The *structural* results use only connectivity and hold regardless of transmitter; the normalizer interpretation we present as a

distributed, evidence-ranked substrate (FB5A its largest and only GABA-predicted member), not a bet on one cell. (FB5A here is the cell type labelled FB5A, distinct from FB5AA/FB5AB.) The *divisive* (shunting) form is a further declared modelling assumption, and the multi-stability anchor is an analogy [Toepfer et al., 2018], not a direct FC2 fit.

The $h\Delta$ selector alternative. As stated in Sec. 4.2, an $h\Delta$ -mediated in-FC2 two-goal selector is not excluded, only bounded: it does not latch at the reference gain but its anatomically-unconstrained gain is unmeasured, and the commitment behaviour coincides with the antipodal regime where $h\Delta$ peaks. Deciding this needs $h\Delta$ physiology.

Method limits. (a) The FC2 preferred bearings ψ_i are inferred from the FC2/PFL columnar fingerprint (not from FB5A or $h\Delta$, so those tests are not circular); the hemibrain uses the real anatomical column index, an independent axis. (b) Irregular real bearings can themselves resist clean latching; we checked this directly with a positive control on the *real* bearings (local excitation injected), which still latches at 78° , above the 60° WTA bound, so a real WTA would be detected, and the negative results are not an artifact of the geometry (though the 78° vs 114° idealized gap confirms the real sampling is somewhat conservative). (c) The distributed substrate is a structural argument; the in-silico model instantiates only FB5A. (d) The decomposition characterizes the routes’ spatial signatures; we do not fit them to reconstruct Mussells Pires et al. [2024]’s suppression-vs-distance curve. (e) FB5A is only four cells, small-population *typing* is fragile; the 88/88 hemibrain replication mitigates the *structural* (coverage) fragility, not the $n=4$ transmitter prediction, which has no independent replication.

7 Conclusion

From a single connectome we decompose the fan-shaped-body goal circuit’s feedback inhibition into a uniform FB5A term and an anti-local $h\Delta$ term, show the wiring cannot host a within-FC2 ring-attractor winner-take-all (tested across five model families, a committed spiking run, and a feedforward sweep, with the wiring replicated in a second connectome), nominate the $h\Delta$ -led recurrent network as the upstream substrate that holds the goal FC2 reads, and exclude the best-modelled alternative (the Lanz $h\Delta$ K-PFG attractor) as its source. FC2 normalizes an externally-set goal rather than selecting one. We interpret FB5A as the APL-type uniform normalizer and specify a two-axis silencing experiment that decides it, alongside the one honestly open alternative, an $h\Delta$ -mediated selector, whose gain is the single measurement that would settle the account.

Code and data availability

All wiring is sourced from the public FlyWire [Dorkenwald et al., 2024, Schlegel et al., 2024] and hemibrain [Scheffer et al., 2020, Hulse et al., 2021] connectomes. All analysis code, the committed result caches, and the test suite are available at <https://github.com/commonorigin/fc2-goal-circuit>; a versioned release is archived at Zenodo with a citable DOI. Every figure and every reported number is produced by a single re-runnable analysis script that writes a committed result cache; a committed test suite checks each reported number against its cache, and every load-bearing number is traced to a claims ledger in the repository. No parameter is fit to any downstream task.

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